

L12b: Why Do Markets Crash?

CHEME 5660 Cornell University

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today, we build a simple agent-based market and ask whether changing one entry of a trading rule can turn a calm market into a falling one.

Today's concepts

- Rules as policies: a lookup table from what neighbors intend to what a trader intends next.
- The dealer: how net buying raises the price, net selling lowers it, and why a switched market needs a control.
- Rule-switch simulation: reading the selling fraction and the price, and what the comparison does and does not show.

What You Need to Know

Core concepts

- A Wolfram rule maps the neighbor sum $c \in \{0, 1, \ldots, 8\}$ to the trader's next intention. Rules 19280 and 19286 differ only at $c = 1$: Buy versus Sell.
- The dealer sets the price from the net trade. We compare every switched market with a control that shares its history.
- Giving the changed rule to a fraction f of the traders can tip the market toward a Sell majority while the control does not. This is an analogy for crashes, not an explanation of them.

For understanding (you won't reproduce these)

- The dealer's price update and the technical traders' band rule.
- The two algorithms, the edge-trader conditions, and the Q-learning notebook.

Lectures and Examples

Lecture:
CHEME-5660-L12b-Lecture-WhyDoMarketsCrash-ABM-Fall-2026.ipynb

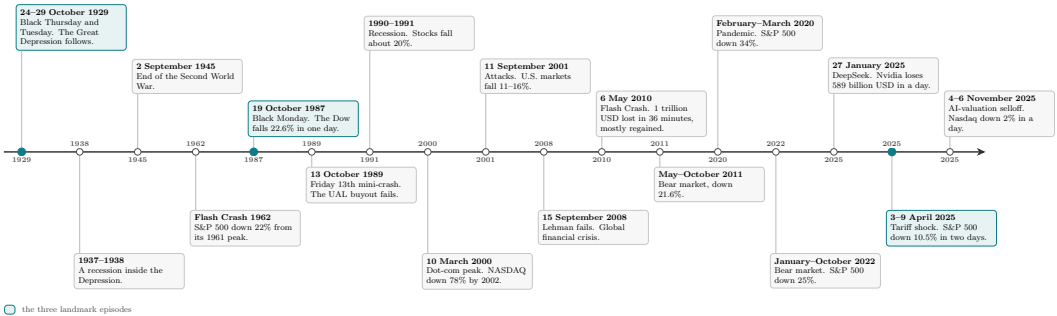
Example:
CHEME-5660-L12b-Example-Wolfram-RuleSwitch-Fall-2026.ipynb

Review on your own:
CHEME-5660-L12b-Example-Wolfram-StylizedFacts-Fall-2026.ipynb

Optional advanced notebooks are listed on the Optional Advanced Material frame.

Why Do Markets Crash?

A market crash is a sudden, significant decline in prices across a broad market. The three teal episodes are described on the next frame:



Three Landmark Episodes

Three episodes, each with a different trigger:

- 1929 Wall Street Crash: margin-fueled speculation met a sudden loss of confidence, and leveraged investors rushed to cover.
- 1987 Black Monday: portfolio insurance and program trading amplified a slide, and a one-day 22.6% Dow drop followed.
- April 2025 tariff mini-crash: tariff announcements and retaliation fears triggered price declines and a jump in volatility. Losses eased after policy clarifications.

Sources: Federal Reserve History; Brady Report; USTR; Federal Reserve FSR

Agent-Based Markets

Markets are man-made systems, and they can fail. An agent-based model simulates traders as autonomous agents with their own rules, and the price emerges from their actions:

- Iori (2002): interacting traders who imitate each other produce large fluctuations and volatility clustering.
- Lux–Marchesi (1999): agents switch between fundamentalist and chartist strategies, producing bubbles and crashes.

Today, traders sit on a grid and act on what their neighbors intend to do.

Sources: Iori; Lux and Marchesi, Nature 397

Agents as Policies

A policy is a table that gives, for each possible state of an agent's surroundings, the action the agent takes. In our market:

- State: a number summarizing what a trader's four neighbors intend.
- Action: the trader's own next intention, Buy, Hold, or Sell.
- Source of the table: Q-learning could learn it from experience. Our traders are handed their tables, and we change one entry.

Wolfram Markets: The Grid and Intentions

Traders sit on the cells of a square grid, and simple local rules produce market-level behavior, as in Wolfram's cellular automata:

- Intention: Buy, Hold, or Sell, coded 0, 1, 2. We write $z_j(t) \in \{0, 1, 2\}$ for the intention of trader j at step t.
- Interior traders: each has four neighbors, above, below, left, and right, and updates its intention from theirs.
- Edge: some edge traders are technical traders who react to the dealer's price. The other edge positions stay in Hold.

Source: our paper on rule switches in Wolfram markets; Wolfram, *A New Kind of Science* (2002), Chapter 3

The Market Model

- Rule: trader i sums the intentions of its four neighbors ∂i, $c_i(t) = \sum_{j \in \partial i} z_j$, and sets $z_i(t) = a^{(R_i)}_{c_i(t)}$, with rule number $\boxed{R = \sum_{c=0}^{8} a_c \, 3^c}$. The baseline is Rule 19280.
- Dealer: quotes $P_t = W_t / Q_t$ (cash over shares), takes the net trade x_t, and keeps $W_t Q_t$ fixed. Depth h is its starting shares per trader:

$$\boxed{P_t = P_{t-1} \Big(\frac{Q_{t-1}}{Q_{t-1} - x_t} \Big)^2}$$

- Technical traders: Buy below the band $\mu_t \pm k_j \sigma_t$, Sell above, Hold inside; active, held, or random signal ($0 < \alpha < 1$):

$$\boxed{\mu_t = \mu_{t-1} + \alpha \, (P_t - \mu_{t-1}) \qquad \sigma_t = \sigma_{t-1} + \alpha \, (|P_t - \mu_t| - \sigma_{t-1})}$$

Algorithm: One Market Step

Each step $t = 1, 2, \ldots, T$ runs in this order, with switch step t_s and R_i the rule of interior trader i:

1. Switch: at $t = t_s$, the chosen fraction f of interior traders get Rule 19286 and keep their intentions, cash, and shares.
2. Clear: the dealer clears the orders from $z_i(t-1)$ and returns x_t, Q_t, W_t, and P_t.
3. Band: update μ_t and σ_t from P_t.
4. Edge: technical traders set $z_j(t)$ from the band (active), Hold (held), or a random draw (random signal).
5. Interior: every interior trader computes $c_i(t)$ from the old interior and new edge intentions, and sets $z_i(t) \leftarrow a_{c_i(t)}^{(R_i)}$.

Full version: the lecture notebook, after Algorithm S1 of our paper

Algorithm: Dealer Clearing

The dealer finds one price $F(x) = W_{t-1}/(Q_{t-1} - x)$ for a net trade x, with excess demand $e(x)$ of buyers who can pay minus sellers with a share:

1. Direction: $d \leftarrow \operatorname{sign} e(0)$, and bounds ℓ, u.
2. Search: bisect for the largest m with $d \cdot e(dm) \geq m$.
3. Price: $x_t \leftarrow d\ell$ and $F_t \leftarrow F(x_t)$.
4. Fill and settle: fill the short side in full; ration the other side by a seeded priority; move cash and shares at F_t.
5. Dealer: $Q_t \leftarrow Q_{t-1} - x_t$, $W_t \leftarrow W_{t-1}Q_{t-1}/Q_t$, and $P_t \leftarrow W_t/Q_t$.

Full version: the lecture notebook, after Algorithm S2 of our paper

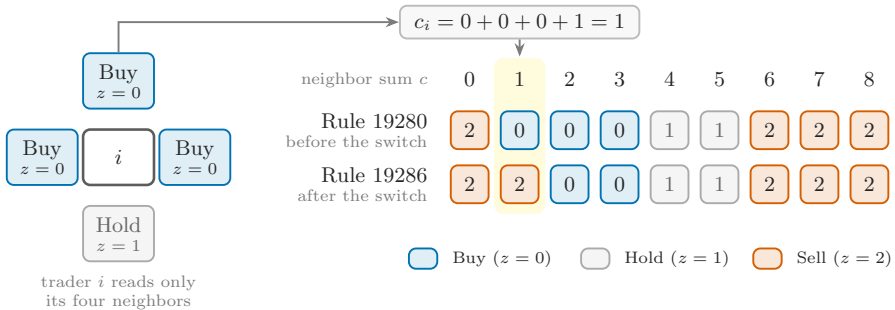
Example: Does the Price Look Real?

Review on your own:
CHEME-5660-L12b-Example-Wolfram-StylizedFacts-Fall-2026.ipynb

In this example, we measure the stylized facts of the dealer's price in a market that never switches, compare them with a real ticker and the Farmer–Joshi market, and vary the edge traders and the dealer's depth.

The Rule Switch

Rule 19286, $(222110022)_3$, differs from the baseline in one entry. Three neighbors intend Buy and one Hold, so $c = 1$: Rule 19280 buys, and Rule 19286 sells. Table columns run from $c = 0$ to 8:



The Rule-Switch Experiment

We use the one-entry difference as a model of a change in psychology. The experiment has three parts:

- Shared history: all interior traders follow Rule 19280. At the switch step, a fraction f of them, chosen in advance, receive Rule 19286 and keep their intentions, cash, and shares.
- Control: a second market with the same history never switches. Differences after the switch come from the rule change.
- Edge traders: active, held, or random signal (a course addition). When held, any cascade comes from traders reading their neighbors.

Example: Rule Switch

Example:
CHEME-5660-L12b-Example-Wolfram-RuleSwitch-Fall-2026.ipynb

In this example, we run a switched market and its control ourselves, compare cached markets under three market conditions, and count how often a Sell majority forms.

Optional Advanced Material

Q-learning: advanced/qlearning/
CHEME-5660-L12b-Advanced-QLearning-Fall-2026.ipynb

How could a trader learn a policy table from experience instead of being handed one? We write the Q-learning update, lay out the algorithm, and describe when it converges to the best policy. Not a prerequisite for L13a.

Summary

We built a simple agent-based market and used it to ask how a small change in trader behavior can tip a market toward selling.

Key takeaways

- Rules as policies: a rule is a lookup table, and one entry can change.
- Dealer and control: net trades set the price, and a control isolates the switch.
- Tipping, by analogy: a switch for some traders can tip the market toward selling, but a simulation does not establish real causes.

Agent-based models ask how individual rules add up to market behavior.